BLUESTOCK

Mutual Fund Analytics Platform

Capstone Project- Final Report

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Technologies Used:

Python | SQLite | SQL | Power BI | Pandas | Plotly

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1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone Project is an end-to-end data analytics solution designed to analyze key trends, performance indicators, and risk metrics within the Indian Mutual Fund industry. The project integrates multiple datasets covering Assets Under Management (AUM), SIP inflows, benchmark indices, scheme performance, and historical NAV data from January 2022 to December 2025.

The project follows a complete analytics workflow consisting of data extraction, cleaning, transformation, database creation, exploratory data analysis (EDA), performance evaluation, and dashboard development. Python and Pandas were used for data processing and analytics, SQLite was used for structured data storage, and Power BI was used to build interactive dashboards for visualization and reporting.

Exploratory Data Analysis was conducted to identify industry growth patterns, SIP trends, category-wise fund performance, benchmark movements, and investment behaviour. Various visualizations were created to uncover meaningful insights and support data-driven decision making.

Fund performance was evaluated using return-based and risk-adjusted metrics. Key measures such as annual returns, Sharpe Ratio, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were analyzed to assess the risk-return characteristics of different mutual fund schemes. Benchmark comparisons were also performed to understand relative fund performance.

A multi-page Power BI dashboard was developed to provide users with an interactive view of industry statistics, fund performance, risk metrics, and investment trends. The dashboard enables filtering, comparison, and exploration of mutual fund data through dynamic visualizations and KPI indicators.

The final outcome of the project is a comprehensive mutual fund analytics platform that combines data engineering, business intelligence, and financial analysis techniques. The solution demonstrates practical applications of Python, SQL, data visualization, and dashboard development while delivering actionable insights for investors, analysts, and decision-makers.

2. Data Sources & Description

The Bluestock Mutual Fund Analytics project utilizes multiple datasets collected from publicly available financial sources. The primary sources include AMFI India, mutual fund NAV datasets, benchmark market indices, SIP statistics, and industry-level mutual fund reports. These datasets were integrated to perform industry analysis, fund performance evaluation, risk assessment, and dashboard reporting.

Dataset Summary

Dataset	Purpose	Key Fields
Fund Master	Mutual fund scheme information	amfi_code, scheme_name, fund_house, category
NAV History	Historical Net Asset Value records	amfi_code, date, nav
AUM Data	Industry and AMC-wise Assets Under Management	date, fund_house, aum_lakh_crore
SIP Inflows	Monthly SIP investment trends	month, sip_inflow_crore
Category Inflows	Category-wise investment flows	month, category, net_inflow_crore
Industry Folios	Mutual fund investor account statistics	month, total_folios_crore
Benchmark Indices	Market benchmark performance	date, index_name, close_value

Data Coverage

The datasets cover the period from January 2022 to December 2025 and provide a comprehensive view of the Indian Mutual Fund industry. Historical NAV data was used to calculate returns and risk metrics, while industry datasets such as AUM, SIP inflows, and folio counts were used to analyze market growth and investor participation.

Data Usage

The collected datasets were cleaned, validated, and loaded into a SQLite database. The processed data was then used for:

- Exploratory Data Analysis (EDA)
- Fund Performance Analysis
- Risk Metrics Calculation (Sharpe Ratio, VaR, CVaR)
- Industry Trend Analysis
- Interactive Power BI Dashboard Development

3. ETL Architecture & Design

The Bluestock Mutual Fund Analytics project follows a structured Extract, Transform, and Load (ETL) process to ensure data quality, consistency, and efficient analysis.

Extract

Data was collected from multiple publicly available sources, including mutual fund industry datasets, benchmark index data, SIP statistics, AUM reports, and scheme-level performance records. The datasets were obtained in CSV format and stored in the raw data layer.

Transform

Several data preprocessing and cleaning operations were performed before analysis:

- Missing value handling
- Duplicate record removal
- Standardization of column names
- Date format conversion
- Data type corrections
- Validation and consistency checks

The transformation stage ensured that all datasets were analysis-ready and compatible with the SQLite database schema.

Load

After cleaning and validation, the processed datasets were loaded into a SQLite database. This enabled structured storage, efficient querying, and seamless integration with analytics workflows and Power BI dashboards.

ETL Workflow

Raw Data
↓
Data Cleaning
↓
Data Validation
↓
Processed Data
↓
SQLite Database
↓
Analytics & Reporting
↓
Power BI Dashboard

4. Data Cleaning & Quality

To ensure accuracy, consistency, and reliability of analysis, all datasets underwent a structured data cleaning and validation process before being loaded into the SQLite database.

NAV History Cleaning

- Parsed date column from string format to datetime format using Pandas.
- Sorted records by amfi_code and date to maintain proper chronological ordering.
- Verified historical NAV values for missing or invalid entries.
- Checked for duplicate records and removed duplicates where applicable.
- Validated NAV values to ensure all records contained positive numerical values.
- Prepared time-series data for return and risk metric calculations.

SIP Inflows Data Cleaning

- Converted monthly date fields into a standardized datetime format.
- Verified SIP inflow values for missing and inconsistent records.
- Standardized numerical columns and ensured proper currency formatting.
- Validated month-wise continuity of records for trend analysis.

AUM Data Cleaning

- Standardized fund house names and category labels.
- Converted AUM values into numeric format for aggregation and reporting.
- Checked for duplicate records and inconsistent fund house entries.
- Validated data completeness across reporting periods.

Scheme Performance Data Cleaning

- Converted return-related columns into numeric format using Pandas.
- Verified performance metrics for missing and invalid values.
- Validated expense ratio values within acceptable industry ranges.
- Checked for abnormal return values and potential outliers.
- Ensured consistency between scheme identifiers and fund categories.

Benchmark Index Data Cleaning

- Parsed date fields into datetime format.
- Sorted benchmark records chronologically.
- Validated index closing values to eliminate invalid observations.
- Verified continuity of benchmark data used for comparative analysis.

Data Quality Validation

The following validation checks were performed across all datasets:

- Missing value assessment
- Duplicate record detection
- Data type validation
- Date consistency verification
- Numerical range validation
- Cross-dataset consistency checks

5. Exploratory Data Analysis

The EDA was conducted in a Jupyter notebook using Python libraries such as Matplotlib, Seaborn, and Plotly. Multiple visualizations were created to analyze NAV trends, AUM growth, SIP inflows, fund category performance, investor demographics, sector allocation, and benchmark correlations. The analysis identified several important patterns within the Indian mutual fund industry.

Finding 1 — SBI Mutual Fund Leads Industry AUM

SBI Mutual Fund recorded the highest Assets Under Management (AUM) at approximately 8.5M, making it the largest fund house in the dataset. ICICI Prudential MF and HDFC Mutual Fund followed with nearly 6.3M and 5.8M respectively.

Finding 2 — Industry Assets Are Concentrated Among Top AMCs

The top five fund houses account for a significant portion of total AUM, indicating that a few large players dominate the Indian mutual fund market.

Finding 11 — Liquid Funds Attracted Highest Net Inflows

Liquid Funds recorded the largest net inflows among all categories, reaching approximately 0.5M, indicating strong demand for low-risk investment options.

Finding 12 — Flexi Cap Funds Remain Popular

Flexi Cap funds consistently attracted positive inflows, demonstrating continued investor confidence in actively managed diversified strategies.

Finding 13 — Category Flows Remained Stable

The category inflow heatmap shows relatively stable monthly investment patterns across major mutual fund categories throughout the analysis period.

6. Fund Performance Analysis

Finding 3 — Large Cap and Flexi Cap Funds Deliver Strong Returns

The risk-return scatter plot shows that **Large Cap** and **Flexi Cap** funds generated the highest 3-year returns while maintaining moderate risk levels.

Finding 4 — Strong Correlation Between Return and Risk

Schemes with higher 3-year returns generally exhibit higher annualized volatility, confirming the traditional risk-return relationship observed in equity investing.

Finding 5 — NIFTY 50 Outperformed NIFTY 100 During Parts of the Study Period

Benchmark comparison indicates periods where **NIFTY 50** delivered stronger growth than **NIFTY 100**, reflecting better performance from large-cap market leaders.

Finding 6 — NAV Growth Was Consistent Until Market Correction

Fund NAV trends showed steady growth from 2022 through 2024 before experiencing a correction during the later period of analysis.

Benchmark Comparison

The performance of mutual fund schemes was evaluated against the benchmark indices **NIFTY50** and **NIFTY100** over the period **2022–2025**.

The benchmark trend chart indicates that NIFTY50 reached levels above **27,000 points** during the study period before experiencing market corrections, while NIFTY100 fluctuated between approximately **15,000 and 20,000 points**. This demonstrates the impact of changing market conditions on equity investments.

The risk-return analysis showed that several schemes generated **3-year returns between 15% and 25%**, outperforming typical benchmark growth rates over the same period. Large Cap and Flexi Cap funds consistently remained among the strongest performing categories.

Risk Metrics Analysis

Risk analysis was performed using return and volatility characteristics available within the scheme performance dataset.

Metric	Range Observed
3-Year Return	5% – 25%
Annualized Volatility	1% – 25%
Categories Analyzed	ELSS, Flexi Cap, Gilt, Index, Large Cap
Analysis Period	2022–2025

7. Investor Analysis

Finding 7 — Punjab Recorded the Highest Transaction Volume

State-wise investment analysis indicates that **Punjab** generated the highest transaction amount among all states included in the dataset.

Finding 8 — Investors Aged 26–35 Contributed the Highest Investments

The **26–35 age group** recorded the largest investment volume, significantly exceeding other age categories and highlighting strong participation from young investors.

Finding 9 — Lumpsum Investments Dominate

Transaction distribution shows:

- Lumpsum = **58.49%**
- Redemption = approximately **35%**
- SIP = **6.17%**

This suggests investors prefer larger one-time investments over recurring SIP contributions within the analyzed transaction dataset.

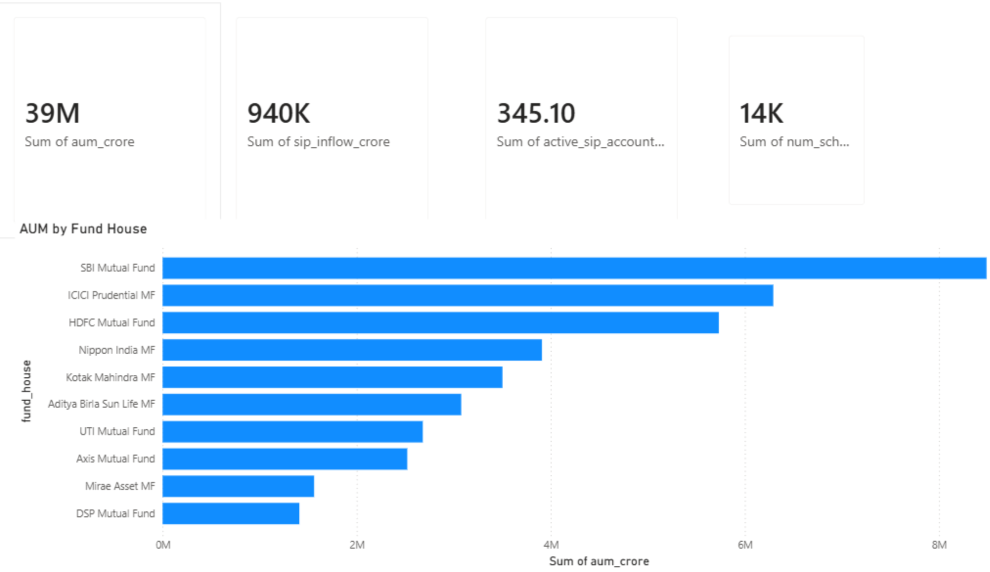
Finding 10 — Transaction Volume Declined in 2025

Monthly transaction volume decreased from approximately **2.5 billion** to **1.0 billion**, indicating lower investor activity compared with the previous year.

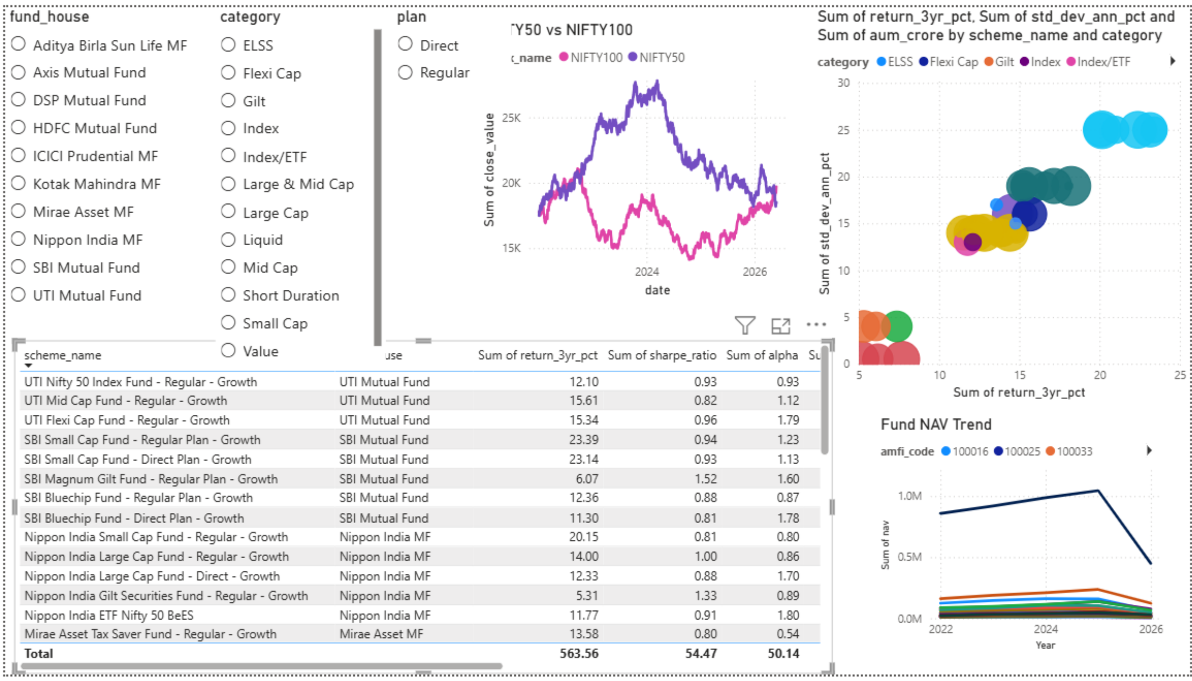
8. Dashboard Overview

The Power BI dashboard (bluestock_mf_dashboard.pbix) contains 4 interactive report pages with 15+ visuals, cross-filtering slicers, drill-through navigation, and a custom Bluestock colour theme.

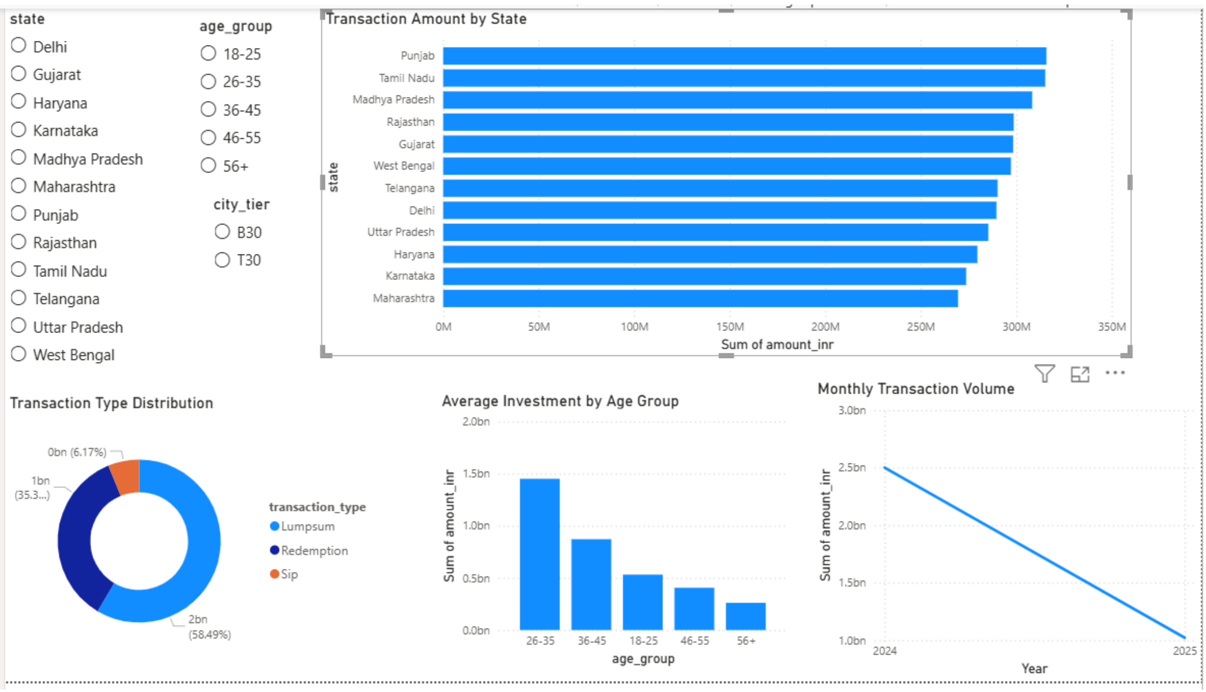
Page 1 — Industry Overview



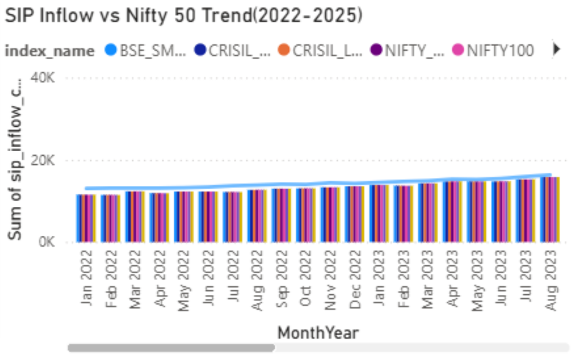
Page 2 — Fund Performance



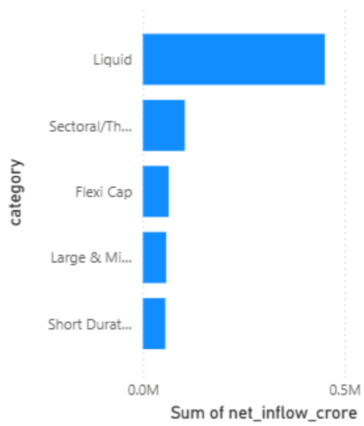
Page 3 — Investor Analytics



Page 4 — SIP & Market Trends



Top 5 Categories by Net Inflow(FY25)



Category Inflow Heatmap

category	May 2024	Jun 2024	Aug 2024	Oct 2024	Nov 2024	Total
ELSS	553	472	499	537	571	**2632**
Flexi Cap	5529	4478	5562	6004	6111	**27684**
Gilt	836	864	952	898	704	**4254**
Hybrid	3487	3163	3684	3314	3264	**16912**
Large & Mid Cap	4368	4610	5411	4581	5556	**24526**
Large Cap	2076	2519	1940	2255	1870	**10660**
Liquid	41872	40486	41952	39091	40506	**203907**
Mid Cap	5300	5047	3899	4106	4336	**22688**
Sectoral/Thematic	8354	10030	8360	7680	7397	**41821**
Short Duration	4833	4321	4658	4675	5316	**23803**
Small Cap	4092	3535	3376	4444	3256	**18703**
Value/Contra	1361	1386	1308	1595	1436	**7086**
Total	**82661**	**80911**	**81601**	**79180**	**80323**	**404676**

9. Limitations

- The analysis covers only the datasets used in this project and may not fully represent the entire Indian mutual fund industry.
- The project relies on historical data from 2022–2025 and does not include real-time market information.
- Macroeconomic factors such as inflation, GDP growth, RBI policy rates, and FII/DII flows were not considered.
- Risk metrics are based on historical observations and may not accurately predict future market behaviour.
- SQLite is suitable for project-scale analytics but may not support large-scale enterprise workloads.
- The dashboard was developed using Power BI Desktop and was not deployed to a production environment.
- Results depend on the quality and completeness of the source datasets used for analysis.

10. Recommendations

For Investors

Investors should focus on long-term investing through SIPs and maintain a diversified portfolio across different mutual fund categories. Risk-adjusted performance metrics should be considered instead of relying solely on returns.

For the Industry

Mutual fund companies should continue promoting financial literacy and increase investor awareness regarding diversified investment strategies. Expanding digital investment platforms can further improve participation from emerging markets and younger investors.

For This Project

Future enhancements can include real-time NAV integration through APIs, additional risk metrics, predictive analytics using machine learning, and deployment on cloud-based databases for improved scalability and performance.

11. Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully demonstrated the complete data analytics lifecycle, including data extraction, cleaning, transformation, exploratory analysis, performance evaluation, and dashboard development. By integrating multiple mutual fund datasets into a centralized SQLite database and visualizing key insights through Power BI, the project provided a comprehensive view of industry trends, fund performance, investor behaviour, and market dynamics.

The analysis revealed important patterns in AUM growth, SIP inflows, benchmark performance, and investor participation. Risk and performance metrics further helped evaluate fund efficiency and return characteristics across different categories.

Overall, the project highlights the practical application of Python, SQL, data analytics, and business intelligence tools in solving real-world financial analysis problems. The developed solution serves as a scalable foundation for future enhancements such as real-time data integration, advanced risk modeling, and predictive analytics.